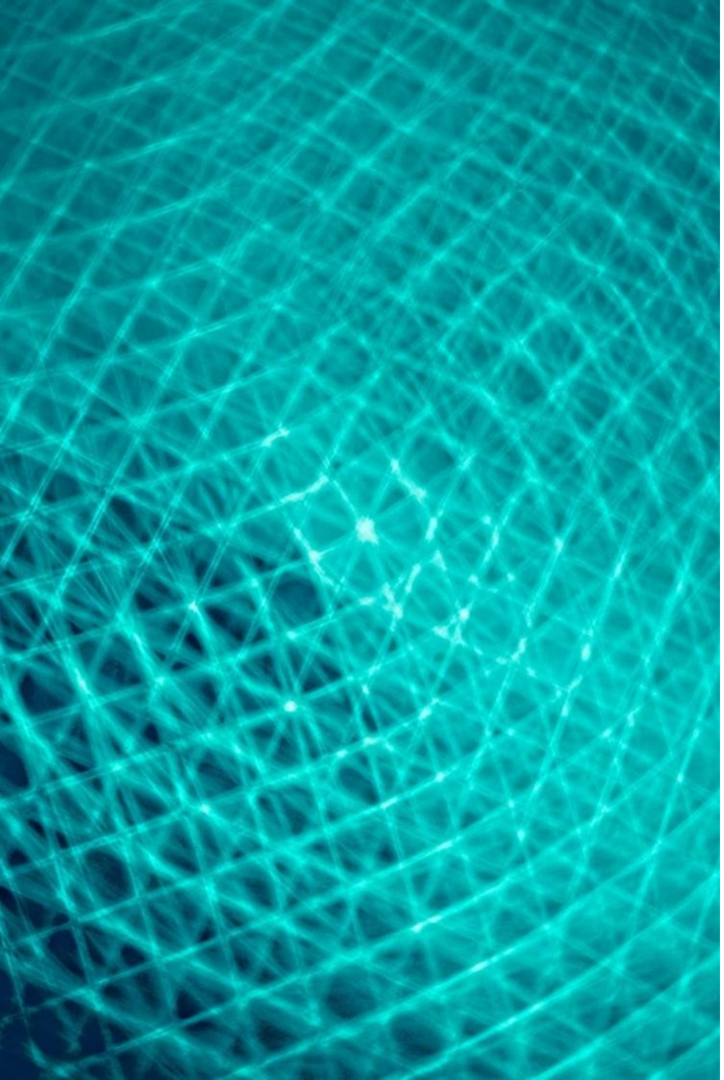

Over-Quantization: Investigating Limits and Error Propagation

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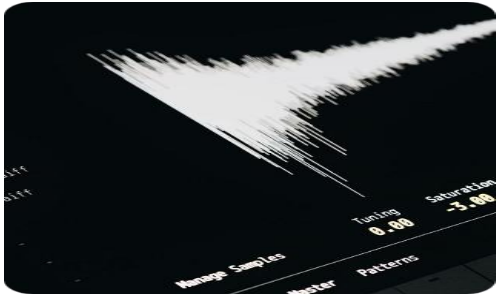
The Quantization Foundation

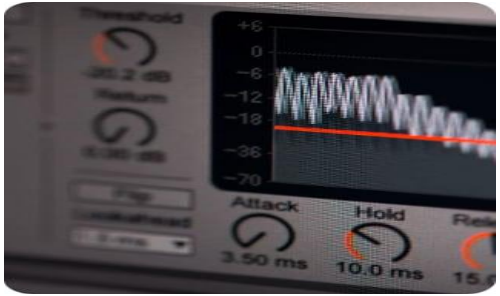
Process

Mapping continuous signals to finite discrete sets (levels or integers). Core trade-off: precision vs. storage/computation.



6 dB Rule

Each additional bit ≈ +6.02 dB in SQNR; practical gains diminish with perceptual masking and system noise floors.



Key Assumption

Quantization noise modelled as uniform, white, and uncorrelated with input — a simplification that breaks in low-bit or structured systems.

When Assumptions Fail

Many theoretical results assume many quantization bins and small step sizes. In practical low-bit regimes, feedback, predictive coding, and modulators introduce correlation between signal and quantization error, producing patterned distortion rather than white noise.

Visualising Information Loss

Low bit-depth creates visible banding (false contouring) as continuous gradations collapse into discrete steps. Reductions in intensity levels force sharp boundaries, losing subtle detail and texture.

Gamut mapping for limited displays/printers further shifts hues and compresses shadow detail, causing detail loss near black points and colour clipping.

The Audio & Signal Threshold

Perceptual Sensitivity

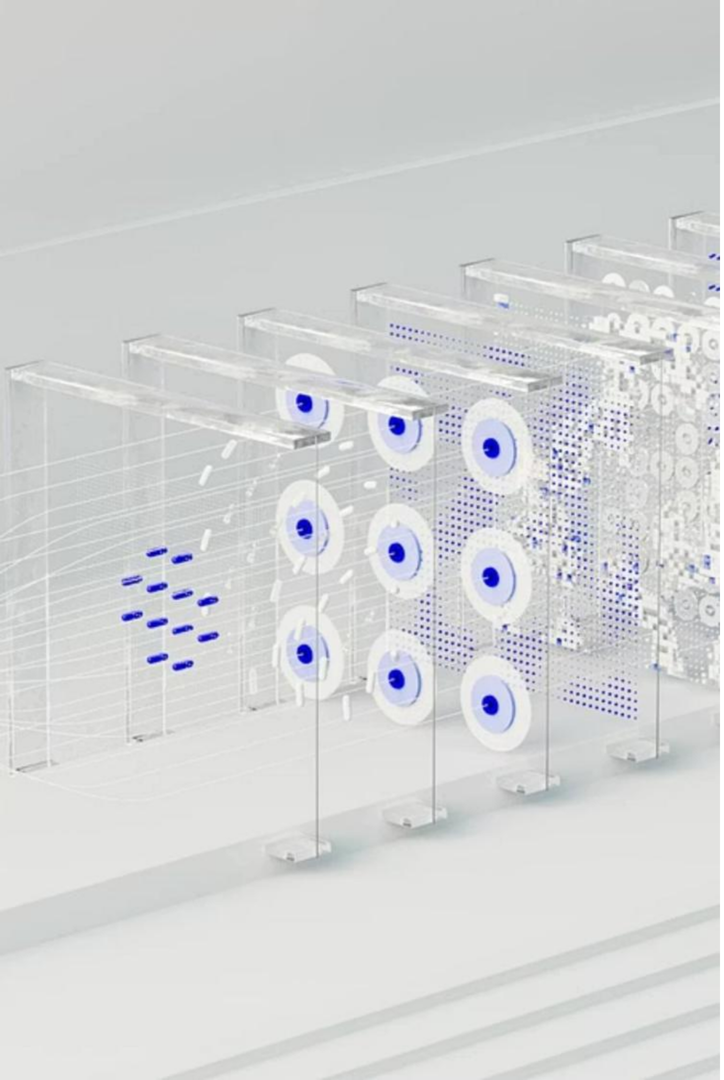
Human hearing is logarithmic; companding (μ-law, A-law) allocates bits to preserve low-level detail.

Dynamic Range

16-bit audio (~98 dB SQNR) usually exceeds environmental noise floors, but processing and resampling can expose quantization artefacts.

Clipping vs Distortion

Analog systems compress gradually; digital clipping produces hard bounds and harmonically-rich distortion that is perceptually intrusive.

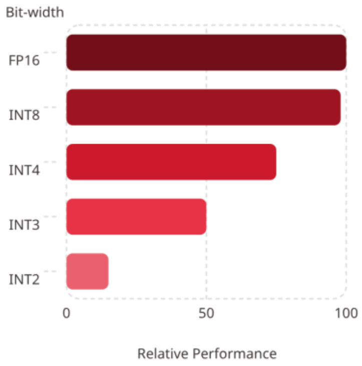


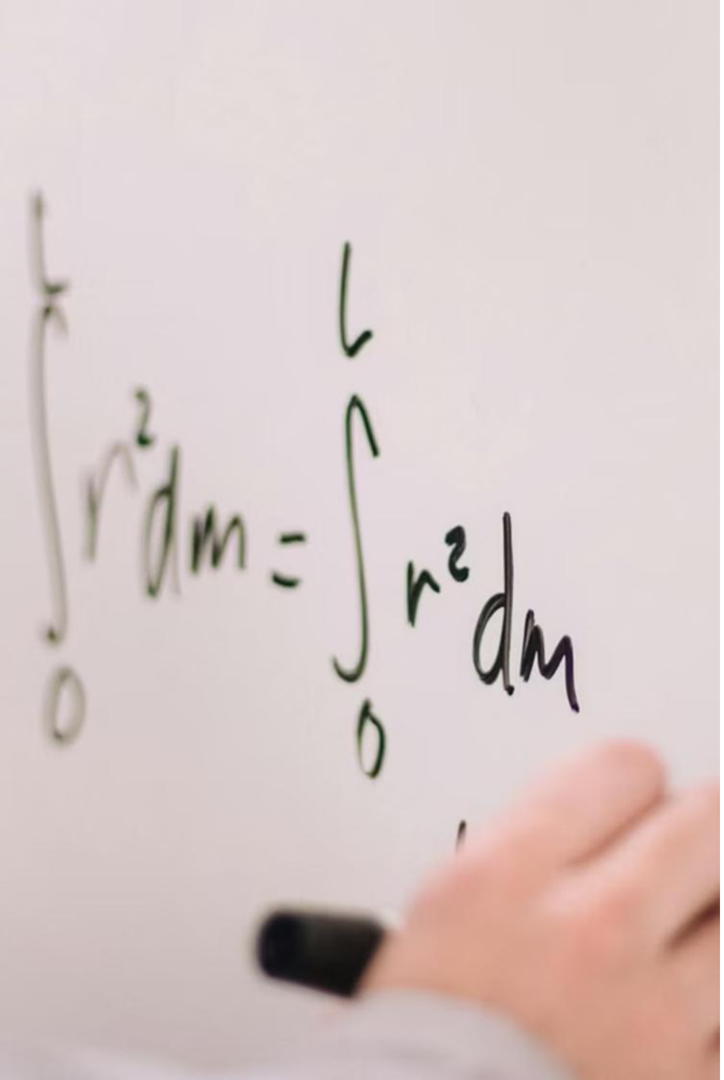
Limitation: Layer-Wise Error Accumulation

Standard post-training quantization treats layers independently. Errors introduced early propagate downstream, interacting with non-linear activations and attention mechanisms — resulting in amplified, structured degradation. In deep transformers, this accumulation often grows near-exponentially.

The Scale of Modern Failure

At INT4/INT3, LLMs show severe accuracy loss; INT2 commonly diverges. Layer-wise calibration often overfits to calibration data, reducing robustness on real inputs. These regimes reveal that naive quantization is not robust for modern large models.





The Solution: Explicit Error Propagation (QEP)

Framework

Quantization Error Propagation (QEP) explicitly tracks upstream errors and inserts corrective terms during layer calibration.

Tunable Strength

Adjustable compensation vs compute trade-off — tuneable to preserve throughput while reducing accumulated distortion.

Compatibility

Designed to integrate with GPTQ, AWQ, QuIP and other PTQ pipelines without disrupting end-to-end deployment.

Prototyping and Performance

Visual demos comparing baseline quantised models versus QEP-enhanced models show reduced error growth and clearer outputs. Benchmarks indicate QEP restores usable accuracy in INT2 regimes where baselines fail.

- Restores task accuracy at very low bit-widths
- Correction terms are computationally lightweight
- Robust across calibration sets — less overfitting

Conclusion: Moving Beyond Uniformity

Recognise Loss

Quantization is inherently lossy — assume structure in errors rather than uniform white noise.

Holistic Calibration

Shift from independent layer optimisation to propagation-aware, system-level calibration.

Future Directions

Adaptive bit-depths, dynamic propagation control, and hybrid correction schemes are promising paths to efficient, high-fidelity deployment.